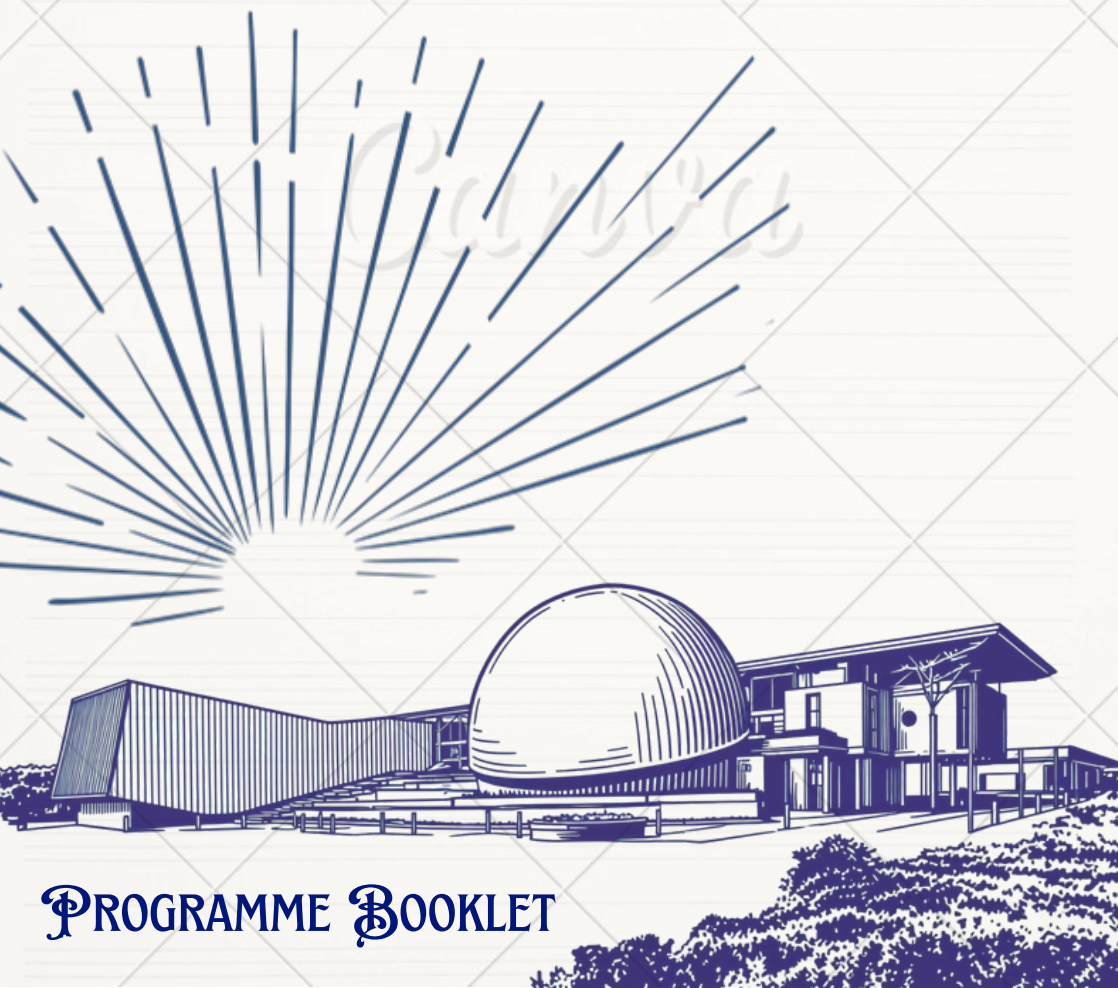




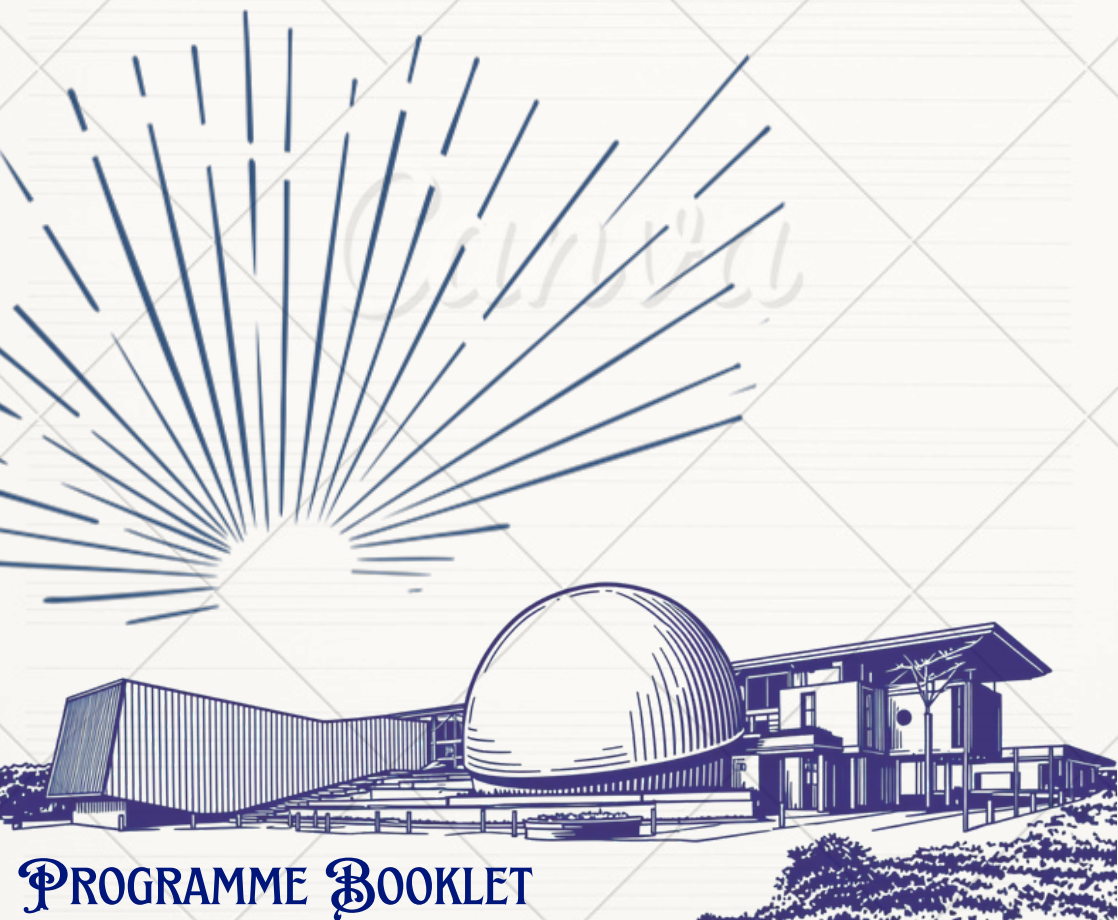
Building An Inclusive Bioinformatics Ecosystem
Nelson Mandela University, Gqeberha
31 August 2026



PROGRAMME BOOKLET



Building An Inclusive Bioinformatics Ecosystem
Nelson Mandela University, Gqeberha
31 August 2026



PROGRAMME BOOKLET

A WARM WELCOME FROM THE ORGANISING COMMITTEE

Dear attendees, colleagues and guests,

It is our distinct privilege to welcome you to the SASBi | SAGS BIO2026 STUDENT SYMPOSIUM. This year, under our umbrella theme, “African Genomes in the Global Village: Landscapes to celebrate diversity” we gather in the beautiful coastal city of Gqeberha (Port Elizabeth) to explore cutting-edge genomic advancements, foster cross-discipline collaborations and celebrate diversity inherent to African genomic landscapes.

This booklet serves as your roadmap for the event, inside you find the session schedules, bios of our esteemed keynotes and panelists, and local insights to help you enjoy your stay.

Thank you for bringing your expertise and passion to this African dialogue.

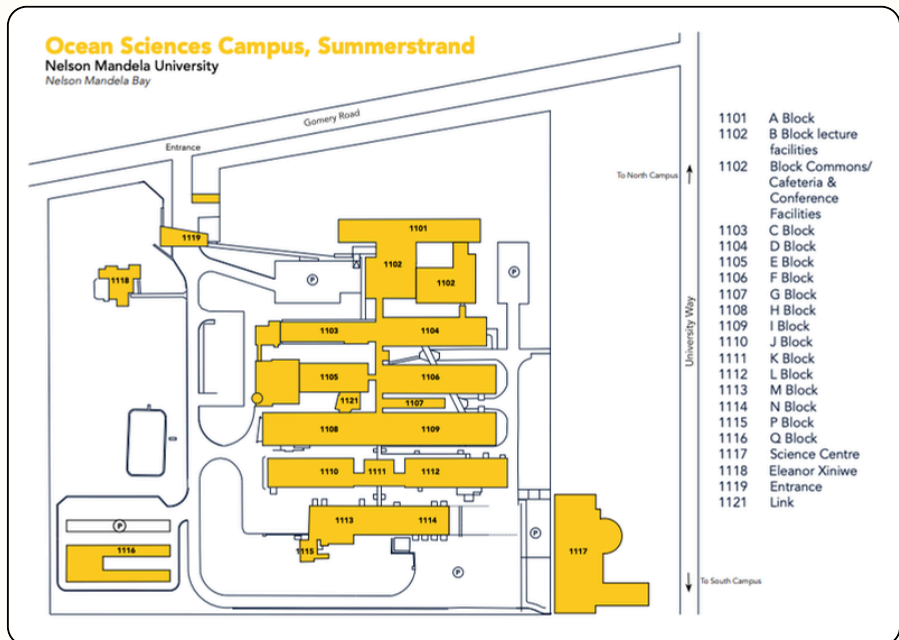
Sincerely,

*Caivil Ndobela and Joshua Sampson
Chairperson, Organising Committee*

VENUE INFORMATION

The SASBi | SAGS BIO2026 Student Symposium will be held at the Digital Dome on Ocean Science Campus, Summerstrand, Nelson Mandela Bay, Nelson Mandela University.

Parking for external guests:



CONNECTIVITY AND DIGITAL ENGAGEMENT

Wi-fi Network Name: Guest

Wi-Fi password: Provided at the venue

LinkedIn: SASBi | SAGS Student Bio-Symposium 2026

Instagram: @student.biosymposium

BlueSky: @biosymposium2026.bsky.social

Official Conference Hashtag: [#BIO26inNMU](#) [#Bioinformatics](#) [#Genetics](#)

EMERGENCY AND HEALTH CONTACTS

On-site first aid:

Report directly to the registration desk or call

Closest Hospital: Life St. George's Hospital.

General Emergency Numbers:

Police: 10111

Ambulance: 10177 or 112

KEYNOTE PRESENTATIONS

Never Stop Evolving: Lessons from a Life in Bioinformatics

Bioinformatics is a field that never stands still and neither can the people working in it. From an unlikely journey into science to a career shaped by constantly evolving technologies, this keynote reflects on the unexpected turns, challenges and opportunities that have shaped one life in bioinformatics. More than a story of where we have come from, it is an invitation to embrace curiosity, adapt to change and never stop evolving.

Dr Rian Pierneef

*Senior Lecturer in Genetics and Bioinformatics, University of Pretoria.
President of the South African Society for Bioinformatics.*

Building the Future of African Bioinformatics: Innovation, Collaboration, and Impact

Bioinformatics in Africa has advanced significantly over the last 2 decades. In addition to capacity developed in individual countries, H3ABioNet had an enormous impact, not only for bioinformatics scientists and users within the consortium, but well beyond the project through its novel and wide-reaching training program. This pan African bioinformatics network built data and computing infrastructure, tools and databases, and most importantly, a community of highly skilled bioinformaticians. However, the scientific landscape in Africa and globally is changing, with the increasing generation of complex datasets and the emergence of evermore sophisticated AI techniques. These provide both challenges and opportunities for our next generation of bioinformatics scientists. In this talk, I will reflect on the trajectory of bioinformatics in Africa, what has been achieved, and how we can address the remaining challenges to make the most of the new opportunities. I will also discuss the role of emerging researchers in shaping the future of bioinformatics on the continent.

Prof Nicola Mulder

Head of the Computational Biology Division at the University of Cape Town (UCT).

PROGRAMME

MONDAY, 31 AUGUST 2026

THEME: AFRICAN GENOMES IN THE GLOBAL VILLAGE: LANDSCAPES TO CELEBRATE DIVERSITY

08:00 - 09:00	REGISTRATION/LOGIN Link: TBC	
09:00 - 09:10	Welcome and housekeeping	Caivil Ndobela, President of SASBI-SC
09:10 - 09:45	Opening Keynote address: Never Stop Evolving: Lessons from a Life in Bioinformatics Dr Rian Pierneef President of the South African Society for Bioinformatics Venue: Auditorium 1	
09:45 - 10:45	Session 1: Oral Presentations	
	Session 1A: Human Landscapes: Population Genomics, Immunity, and health diversity Focus: Cohort genomics, host immune responses, and disease susceptibility within African and diaspora populations Facilitator: Odireleng Mosuwe Co-Facilitator: Nozipho Magagula Venue: Auditorium 1	Session 1B: Pathogen Landscapes: Vector Biology, Host-Parasite Dynamics, and Veterinary Genomics Focus: Surveillance of agricultural, veterinary, and vector-borne pathogens alongside innovative biological and chemical control methods Facilitator: 1 OC Member Co-Facilitator: OC Member Venue: Auditorium 2
09:45 - 10:00	Phylogenetic age is associated with mitochondrial heteroplasmy in African and diaspora populations - Dayna Adrienne Croock	Characterisation of the Subcellular Localisation of African Horse Sickness Virus Non-structural Protein NS5 - Nipho Mtambo
10:00 - 10:15	CCR5 regulatory variants and novel loci associated with susceptibility to HIV-1 infection in Sub-Saharan African populations - Isabel du Randt	Genetic Diversity of Ehrlichia Ruminantium Based on Multilocus Sequence Typing of Isolates from Indigenous Goats and Amblyomma Hebraeum Ticks - Xolile Nuse
10:15 - 10:30	Sex-Specific Gene Expression Signatures as Biomarkers in predicting Tuberculosis Relapse - Bongani Mnyandli	Genomic Surveillance of Trehalose-6-Phosphate Synthase in Anopheles Gambiae: A Promising target for Next-Generation Insecticides - Mercy Bella-Omunagbe

10:30 - 10:45	Transcriptomic Characterisation of Differentiation-driven Priming of the Interferon- α Response in THP-1 macrophages - Boithuso Phale	Isolation and Characterization of Biocide Producing Fungus Against Anopheles Mosquitoes from Pesticide Factory Effluents - Tariku Feyisa Wakjira
10:45 - 11:10	Tea Break	
11:15 - 12:15	Panel discussion: Building an inclusive and equitable bioinformatics ecosystem Session Chair: Joshua Sampson, President of the SAGS-Student Society Panel Speakers: Prof. Scott Hazelhurst (), Prof. Tastan-Bishop (), Lebogang Siviya(), Mr. Siyanda Mazibuko (), Prof. Carminita Frost () Venue: Auditorium 1 Link: TBC	
12:15 - 13:25	Picture Moment #BIO26inNMU #Bioinformatics #Genetics	
12:25 - 13:10	Lunch	
13:15 - 14:15	Session 2: Oral Presentations	
	Session 2A: Computation Landscapes: Machine Learning, Bioinformatic Pipelines, and Data Discovery Focus: Advanced algorithmic approaches, neural networks, automated screening workflows, and sequence classifiers applied to oncology and genomic tracking Facilitator: Odireleng Mosuwe Co-Facilitator: OC member Venue: Auditorium 1	Session 2A: Ecological Landscapes: Wildlife conservation, Environmental Health, and Ecosystem Diversity Focus: Tracking diversity across diverse biomes, spanning African wildlife conservation, river systems, aquaculture and agricultural ecology Facilitator: Caivil Ndobela Co-Facilitator: Nozipho Magagula Venue: Auditorium 2
13:15 - 13:30	Graph Neural Network-Based Discovery of miRNA-mRNA Regulatory Interactions in Breast Cancer - Nothando Gama	AMR in Freshwater Environments of the Okavango Delta and Ngamiland Regions - Patience Motshosi
13:30 - 13:45	Reconstructing Cervical Cancer Tumour Microenvironment States from DNA Methylation Profiles Using a Machine Learning Framework - Saltiel Hamese	Assessing the Genetic Diversity and Structure of Spotted Hyena (<i>Crocuta Crocuta</i>) populations across Africa - Kagisho Dibakoane
13:45 - 14:00	DockVina: An Automated Molecular Docking Pipeline for High Throughput Virtual Screening - Siphamandla Mnakeli Dlamini	Temporal Dynamics of the Gut Microbiome During Smoltification in Atlantic Salmon (<i>Salmo Salar</i>) - Muktaar Badrorin

14:00 - 14:45	Taxonomic Binning of Long-Read DNA Sequences by K-mer Composition using Decision Tree Classifiers - Bruhan Kyomuhendo	Unravelling the Elusive Mating Strategy of <i>Dematophora Necatrix</i> : A Combined Genomic and Physiological Approach - Annabel Norval
14:15 - 14:25	Tea Break	
14:30 - 15:30	Session 3: Flash talks Session Chair: Joshua Sampson , President of SAGS SC and Sandile Ntloko , Deputy Development of SASBI-SC	
14:30 - 14:37	Advancing avian genomics: A de novo assembly pipeline to generate a high-quality reference genome for the Southern Bald Ibis (<i>Geronticus calvus</i>) - Ashley Colleen Mortimer	
14:37 - 14:44	An Immunoinformatics Approach for Cross-Species Diagnostics: Validating a <i>Fasciola gigantica</i> Surrogate Antigen for <i>Schistosoma haematobium</i> Detection - Adedayo Adesida	
14:44 - 14:51	Comparative in silico prediction and characterisation of secondary metabolites of <i>Exserohilum Turcicum</i> and <i>Exserohilum Rostratum</i> - Catherina Kuetezang	
14:51 - 14:58	Does multi-task supervision teach a classifier where to look? A quantitative attention-localization study in brain tumor MRI - Tisetso Letuka	
14:58 - 15:05	The Role of the Gut Microbiome in Inflammation and Multimorbidity in African Populations: Evidence from AWI-GEN - Aaron Berkman	
15:05 - 15:12	Impact of Reference Genome, Alignment Strategy, and Quantification Method on Locus-Specific Human Endogenous Retrovirus (HERV) Detection During Monocyte-to-Macrophage Differentiation - Tebogo Phago	
15:12 - 15:19	Integrated Phenotypic Profiling and Therapeutic Target Prioritization of Clinical Multidrug-Resistant <i>Pseudomonas aeruginosa</i> Exhibiting Biofilm-Driven Persistence - Sinethemba Yakobi	
15:19 - 15:26	Benchmarking DNBSEQ-T7 Sequencing Performance Using Genome in a Bottle Reference Genomes: Impact of Library Preparation Strategies - Maano Malima	
15:26 - 15:30	Session Wrap Up Q&A Session Chair: Nozipho Magagula , Deputy secretary & Treasurer of SASBI-SC	

15:30 - 16:00	Closing Keynote address: Building the Future of African Bioinformatics: Innovation, Collaboration, and Impact Prof Nicola Mulder Head of the Computational Biology (CBIO) division at the University of Cape Town, Principal investigator of H3ABioNe, and ABI Founding member Venue: Auditorium 1 Link: TBC	
16:00 - 16:30	Sponsors and Prize giving	
16:30 - 17:00	Annual general meeting (AGM) SASBi Session chair: Caivil Ndobela, President of SASBi SC Venue: Auditorium 1 See agenda here.	Annual General Meeting (AGM) SAGS Session chair: Joshua Sampson, President of SAGS SC Venue: Auditorium 2 See agenda here.

MEET THE KEYNOTE SPEAKERS

OPENING KEYNOTE SPEAKER



Dr Rian Pierneef

Senior Lecturer in Genetics and Bioinformatics, University of Pretoria.

Dr. Rian Pierneef is a Senior Lecturer in the Department of Biochemistry, Genetics and Microbiology at the University of Pretoria and is affiliated with the Centre for Bioinformatics and Computational Biology (CBCB). His teaching and research focus on bioinformatics, genomics, and computational biology, with particular interests in microbial genomics, metagenomics, transcriptomics, and the development of reproducible bioinformatics workflows for next-generation sequencing data analysis. He is passionate about interdisciplinary research and the application of computational approaches to address challenges in human health, agriculture, and environmental microbiology. Dr. Pierneef is dedicated to mentoring students and advancing bioinformatics education and research in South Africa.

CLOSING KEYNOTE SPEAKER



Prof Nicola Mulder

Head of the Computational Biology Division at the University of Cape Town (UCT).

Prof Mulder heads the Computational Biology Division at UCT. She is PI of the DS-I Africa eLwazi Open Data Science Platform and co-leads a Sickle Cell Disease Data Coordinating Centre and a Wellcome Trust Discovery Data Integration Platform at UCT. She led H3ABioNet, a Pan African Bioinformatics Network which developed bioinformatics capacity to enable genomic data analysis for more than a decade. Her research focuses on genetic determinants of susceptibility to disease, African genome variation, and microbial genomics and infectious diseases. Her group provides bioinformatics services and training and develops new algorithms and resources for African genomics. Prof Mulder is actively involved in capacity development in Bioinformatics and genomics. She also sits on a number of international scientific advisory boards and is currently the interim Director for the African Bioinformatics Institute.



Prof Carminita Frost

Professor of Biochemistry at Nelson Mandela University.

Carminita Frost is a Professor of Biochemistry at Nelson Mandela University (NMU, Gqeberha, South Africa) and a scientist in bioprospecting and phytomedicine translation. Specializing in the biochemical evaluation and functional validation of indigenous South African flora, her research platform bridges traditional ethnobotanical knowledge with modern drug discovery. Prof. Frost's current program focuses on biotech innovation in the medicinal cannabis sector, where her group identifies bioactive pipelines and elucidates their molecular mechanisms of action against chronic metabolic and targets, including diabetes and cancer. Notably, her laboratory is work on Cannabis sativa, investigating novel, synergistic cannabinoid formulations engineered to induce programmed cell death in breast cancer lines via paraptosis—a unique, non-apoptotic pathway with significant therapeutic potential for overcoming drug resistance. An active leader in the scientific community, Prof. Frost has served on the South African Society of Biochemistry and Molecular Biology (SASBMB) council and maintains a robust network of national and international academic and industry collaborations. The work completed in her group includes publications and she enjoys mentoring the next generation of biotech talent.



Prof Özlem Tastan-Bishop

Full Professor of Structural Bioinformatics at Rhodes University, South Africa, and a Distinguished Adjunct Professor at Saveetha University, Chennai, India.

She obtained her BSc degree in Physics from Boğaziçi University, Istanbul, Turkey, before completing an MSc in Molecular Biology and Genetics at the same institution. She received her PhD in 2003 from the Max Planck Institute for Molecular Genetics and the Free University of Berlin, Germany. During her doctoral studies, she developed a strong interest in structural biology, which she further pursued through postdoctoral appointments at the University of Texas, USA, the University of the Western Cape, and the University of Pretoria, South Africa, where she gained expertise in structural biology and structural bioinformatics. In 2009, Prof. Tastan Bishop joined Rhodes University as an academic staff member. In 2013, she founded the Research Unit in Bioinformatics (RUBI), which has since become a leading research group in structural bioinformatics.

Since joining Rhodes University, she has supervised and graduated 29 PhD and 42 MSc students. In recognition of her research excellence, she received the Rhodes University Vice-Chancellor's Distinguished Senior Research Award in 2020 and the South African Society for Bioinformatics (SASBi) Silver Award in 2022.

Prof. Tastan Bishop is a Fellow of the Royal Society of South Africa (FRSSAf), where she serves as a Council Member and Eastern Cape Representative. She is also a Member of the Academy of Science of South Africa (MASSAf) and an Associate of the National Institute for Theoretical and Computational Sciences (NITeCS). She serves on the Editorial Boards of PLOS ONE, Frontiers in Molecular Biosciences, and Frontiers in Applied Mathematics and Statistics (Biological Modeling and Simulation section). In addition, she has acted as a reviewer for numerous leading international peer-reviewed journals.

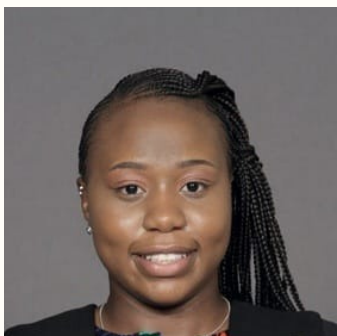
Her research focuses on structural bioinformatics and its applications to drug discovery and development. Her recent work centres on understanding protein allostery, elucidating the structural and functional effects of nonsynonymous single nucleotide variants, and integrating artificial intelligence with structural bioinformatics to advance drug discovery and pharmacogenomics. She has published over 100 peer-reviewed research articles.



Prof Scott Hazelhurst

Professor of Bioinformatics in the School of Electrical and Information Engineering at Wits.

Scott Hazelhurst is professor of bioinformatics in the School of Electrical and Information Engineering and Senior Scientist at the Sydney Brenner Institute for Molecular Bioscience at the University of the Witwatersrand, Johannesburg. He is Co-PI of the Data Science for Health Discovery and Innovation in Africa MADIVA Research Hub, and a co-investigator on the eLwazi Open Data Science Platform, and collaborated on several H3Africa projects, co-leading the AWI-Gen Microbiome Project. His areas of interests include bioinformatics, health informatics and high performance computing. Scott received his honours and MSc degrees from Wits and his PhD from the University of British Columbia.



Ms Lebogang Siviya

PhD Candidate, Nelson Mandela University.

Lebogang Siviya is a PhD candidate in Botany at Nelson Mandela University, where her research focuses on using multi-omics approaches to understand the molecular responses of sweet potato to abiotic stress. She holds an MSc in Molecular and Cell Biology (with distinction) from the University of the Witwatersrand, where she investigated virus-host interactions in tomatoes. Her work aims to contribute to crop improvement and food security by advancing our understanding of the molecular mechanisms underlying crop adaptations to the environment. Her research interests include plant biology, biotechnology and bioinformatics with a focus on integrating these disciplines to address agricultural challenges. Beyond her research, Lebogang is actively involved in mentoring students in molecular biology, genetics, plant biotechnology and research.



Mr Siyanda Mazibuko

Associate Lecturer, Nelson Mandela University.

Siyanda Mazibuko is an Associate Lecturer in the Department of Biochemistry and Microbiology at Nelson Mandela University. His MSc research focuses on antimicrobial resistance (AMR) surveillance in wastewater using molecular microbiology, microbial genomics, and bioinformatics. He employs PCR, Sanger sequencing, and Oxford Nanopore sequencing to investigate antimicrobial resistance determinants in clinically relevant bacteria. Prior to his current appointment, Siyanda served as a Postgraduate Scholarship and Ethics Administrator within Nelson Mandela University's Department of Research Development. He also completed specialised training in digital PCR through the Centre for Epidemic Response and Innovation (CERI) and QIAGEN, strengthening his expertise in molecular diagnostics and genomic technologies. In addition to his research, Siyanda teaches undergraduate biochemistry and has been recognised for his research presentations at both university and departmental level, including

receiving an award in the university wide Three Minute Thesis (3MT) competition. Through his research, teaching, and continued professional development, he is committed to advancing genomic technologies to strengthen antimicrobial resistance surveillance and improve public health.

PRESENTATION ABSTRACTS

PHYLOGENETIC AGE IS ASSOCIATED WITH MITOCHONDRIAL HETEROPLASMY IN AFRICAN AND DIASPORA POPULATIONS

Dayna Adrienne Croock*, Elouise Kroon¹, Devon Allie¹, André G. Loxton¹, Marlo Möller^{1,2,3,4}, Caitlin Uren^{1,2,3,4}, Desirée C. Petersen^{1,2,3,4}

¹South African Medical Research Council (SAMRC) Centre for Tuberculosis Research (CTR), Division of Molecular Biology and Human Genetics (MBHG), Department of Biomedical Sciences, Faculty of Medicine and Health Sciences, Stellenbosch University, Cape Town, South Africa, ²Centre for Bioinformatics and Computational Biology, Stellenbosch University, Cape Town, South Africa, ³Genomics for Health in Africa (GHA), Africa-Europe Cluster of Research Excellence (CoRE), South Africa, ⁴National Institute for Theoretical and Computational Sciences (NITheCS), South Africa, *dayna@sun.ac.za

Background: Mitochondrial DNA (mtDNA) heteroplasmy—the co-existence of multiple mtDNA variants within an individual—has been linked to ageing, disease susceptibility, and cellular dysfunction. However, the evolutionary and population-level factors governing heteroplasmy burden remain poorly understood. Recent work in green sea turtles demonstrated that individuals carrying phylogenetically derived mtDNA haplotypes accumulate more heteroplasmic variants than those carrying ancestral haplotypes, suggesting that lineage age may influence heteroplasmy burden. Here, we test whether a similar phylogenetic age effect exists in humans using whole mitochondrial genome sequencing data from 1,306 individuals representing African and diaspora populations. Using negative binomial and Poisson generalised linear models with L0—the most ancestral human mtDNA macrohaplogroup—as the reference, we find that carriers of L1, L2, and L3 haplogroups exhibit significantly higher heteroplasmy counts (incidence rate ratios 2.71–3.41, all FDR-adjusted $p < 10^{-39}$). This pattern persists after controlling for broad geographic region and is replicated in a diaspora-only subset with shared demographic history, suggesting that the association is not solely explained by population structure. Importantly, using calibrated sub-haplogroup age estimates, we identify a significant negative relationship between sub-haplogroup age and heteroplasmy burden within L0 itself (IRR = 0.978 per thousand years, $p = 3.8 \times 10^{-14}$). Each additional 1,000 years of lineage age is associated with approximately 2.2% fewer heteroplasmic variants. This within-lineage gradient, independent of between-haplogroup differences, provides strong evidence that phylogenetic age contributes to shaping mtDNA heteroplasmy burden in humans, suggesting that long-term evolutionary history influences contemporary patterns of mitochondrial variation.

TRANSCRIPTOMIC CHARACTERISATION OF DIFFERENTIATION-DRIVEN PRIMING OF THE INTERFERON-A RESPONSE IN THP-1 MACROPHAGES

Boithuso Phale^{1*}, Prumohdya D. Muamba¹, Vanessa Meyer¹, Nikki L. Gentle¹

¹School of Molecular and Cell Biology, University of the Witwatersrand, 2451444@students.wits.ac.za

Monocytes and macrophages are important coordinators of the innate response to infectious disease. During infection, monocytes are recruited to tissues where they differentiate into macrophages, which mount interferon (IFN)-driven antiviral responses through the expression of IFN-stimulated genes (ISGs). While ISG expression is classically attributed to IFN signalling, the contribution of monocyte-to-macrophage differentiation to priming this response remains poorly understood. The aim of this study was to investigate how differentiation primes the IFN response in macrophages, using THP-1 cells as a model system. To define the macrophage IFN response, differential gene expression (DGE) was performed with DESeq2 on RNA-sequencing data from THP-1 macrophages differentiated with PMA for 24 hours and stimulated with IFN- α 2 for four hours. This identified 978 significantly differentially expressed ISGs ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$). To assess ISG dynamics during differentiation, Cap Analysis of Gene Expression sequencing (CAGE-seq) data spanning 96 hours of THP-1 monocyte-to-macrophage differentiation were analysed using CAGEfightR. Promoters were defined as transcription start site clusters supported in at least two samples within 20 bp, while enhancers were defined as bidirectional clusters supported in at least two samples. Differential expression analysis of promoter- and enhancer-associated genes revealed progressive ISG induction, increasing from 24 differentially expressed ISGs during early differentiation to 173 ISGs by 24 hours without cytokine stimulation. Hierarchical clustering of ISG expression revealed distinct temporal patterns, including early-induced and sustained, late-induced, and transiently induced ISGs. Motif-enrichment analysis with motifmatchr and JASPAR identified EGR1, EGR2, SP1, and IRF8 as potential key regulators of this ISG priming. Together, these findings suggest that differentiation progressively rewires ISG expression through distinct regulatory waves that prime macrophages for a robust IFN response.

CCR5 REGULATORY VARIANTS AND NOVEL LOCI ASSOCIATED WITH SUSCEPTIBILITY TO HIV-1 INFECTION IN SUB-SAHARAN AFRICAN POPULATIONS

Isabel du Randt^{1,2,3,*}, Wenlong C. Chen^{1,2,7,21}, Irene Namara^{4,5,6}, Luicer A. Ingasia Olubayo¹, Alexander J. Mentzer^{13,14}, Mazvita Muchengeti^{7,8,9,21}, Chantal B. de Villiers³, Rob Newton^{10,14}, Freddy Sitas^{15,16,17}, Debbie Bradshaw¹⁵, Tim Waterboer²⁰, June Fabian^{18,19}, Segun Fatumo^{4,18,21}, Michèle Ramsay¹³, Christopher G. Mathew^{13,21}, Jean-Tristan Brandenburger^{1,2,21}

¹Sydney Brenner Institute for Molecular Bioscience, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa, ²Strengthening Oncology Services Research Unit, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa, ³Division of Human Genetics, School of Pathology, NHLS and Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa, ⁴MRC/UVRI and LSHTM Uganda Research Unit, Entebbe, Uganda, ⁵Makerere University, Kampala, Uganda, ⁶Institute of Translational Genomics, Helmholtz Zentrum München, German Research Center for Environmental Health, 85764 Neuherberg, Germany, ⁷National Cancer Registry, a Division of the NICD, NHLS, Sandringham, Johannesburg, South Africa, ⁸School of Public Health, University of the Witwatersrand, Johannesburg, South Africa, ⁹South African DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA), Stellenbosch University, Stellenbosch, South Africa, ¹⁰Medical Research Council/ Uganda Virus Research Institute/London School of Hygiene and Tropical Medicine (MRC/UVRI/LSHTM) Uganda Research Unit, Entebbe, Uganda, ¹¹The Department of Non-communicable Disease Epidemiology, London School of Hygiene and Tropical Medicine, London, UK, ¹²The Centre for Human Genetics, University of Oxford, Oxford, UK, ¹³Chinese Academy of Medical Science Oxford Institute, University of Oxford, Oxford, UK, ¹⁴University of York, YO10 5DD York, UK, ¹⁵Burden of Disease Research Unit, South African Medical Research Council, Cape Town 7595, South Africa, ¹⁶International Centre for Future Health Systems, University of New South Wales, Sydney Australia, ¹⁷School of Population Health, University of New South Wales, Sydney, Australia, ¹⁸Medical Research Council/Wits University Rural Public Health and Health Transitions Research Unit (Agincourt), School of Public Health, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, Gauteng, South Africa, ¹⁹Wits Donald Gordon Medical Research Institute, Faculty of Health Sciences, University of Witwatersrand, Johannesburg, Gauteng, 2193, South Africa, ²⁰Infections and Cancer Epidemiology, German Cancer Research Center (DKFZ), Heidelberg, Germany, ²¹Network for Oncology Research in Africa (NORA), Global Health Working Group, Martin-Luther-University, Halle-Wittenberg, Germany *fouriesabel1@gmail.com

Background: Human immunodeficiency virus type 1 (HIV-1) infection and sequelae remain a major global health challenge, with Sub-Saharan Africa (SSA) bearing the highest disease burden. Despite evidence that host genetic factors influence HIV-1 susceptibility, most genome-wide association studies (GWAS) have been conducted in non-African populations, leaving the genetic basis of susceptibility in SSA largely unexplored. High HIV-1 prevalence in SSA increases the likelihood that controls were exposed but uninfected, improving the interpretability of case-control analyses compared with studies in low-prevalence populations. **Methods:** We conducted the largest GWAS to date of susceptibility to HIV-1 infection in individuals of SSA ancestry, including 3,069 HIV-1 positive cases and 14,291 HIV-1 negative controls, from multiple study populations from South and East Africa. Fine-mapping and colocalisation analysis with blood expression quantitative trait loci (eQTLs) from the Southern African Blood Regulatory (SABR) resource were used to evaluate functional mechanisms underlying associated loci. **Results:** Three independent loci reached genome-wide significance for HIV-1 infection: rs20409388 and rs2227010 on chromosome 3 (in CCR5AS and upstream of CCR5), rs112358207 on chromosome 1 (in ESRRG), and rs113298445 on chromosome 8 (in INTS9). The chromosome 3 signal colocalised with eQTL variants for CCR5AS expression. Although the protective CCR5Δ32 mutation is absent in African populations, our findings identified alternative variants in the CCR5 regulatory region that may modulate susceptibility to HIV-1 infection. **Conclusions:** This study revealed both shared and population-specific genetic determinants of susceptibility to HIV-1 infection, underscoring the central role of CCR5 regulation in host defence. By expanding genomic studies to African populations, we identify novel loci and refine our understanding of HIV-1 infection in the populations most affected by the epidemic. Further replication and functional analyses are needed to establish causal mechanisms and evaluate translational relevance.

SEX-SPECIFIC GENE EXPRESSION SIGNATURES AS BIOMARKERS IN PREDICTING TUBERCULOSIS RELAPSE

Bongani Mnyandli¹*, Anandi Bierman¹, Gerhard Wazi¹

¹Stellenbosch University, Stellenbosch, South Africa, *nzmandebongani95@gmail.com

Background: Tuberculosis (TB) relapse, defined as the recurrence of active disease after treatment completion, is a major challenge in the treatment of the disease. The lack of reliable biomarkers, along with the sex-dependent biological differences in disease. Highlight the need for the identification of minimal, sex-specific transcriptional signatures for TB relapse to guide treatment monitoring strategies. **Methods:** We analysed gene expression data from the PREDICT-TB cohort, employing a robust three-stage machine learning feature selection pipeline (filtering, feature importance, and wrapper optimization) on the male subset of the dataset. The final minimal gene panel was evaluated using Random Forest (RF) and eXtreme Gradient Boosting (XGBoost) classifiers, with metrics such as using metrics including Area Under the Curve (AUC), Sensitivity, and Specificity. Performance was assessed via repeated k-fold cross-validation and subsequently tested on independent external validation cohorts. **Results:** We successfully identified a 7-gene signature (ENTPD2, KANK1, ANAPC13, PI4K2A, INTS1, ZNF439 and NFATC2IP-AS1) capable of discriminating relapsed from cured patients in the internal cross-validation set. The RF model yielded the highest predictive performance (AUC = 0.9656), validating the concept of a potent sex-specific predictive signature. However, predictive performance declined significantly when tested on external validation cohorts, indicating a failure of generalizability to the broader population. **Conclusion:** The 7-gene signature serves as a strong mechanistic hypothesis for male-specific relapse susceptibility, identifying biological pathways that may be therapeutically targetable. The failure of the model to generalize highlighted its sensitivity to cohort-specific batch effects. Future work must prioritize cross-cohort data harmonization and the development of a signature for the female population for clinical translation.

CHARACTERISATION OF THE SUBCELLULAR LOCALISATION OF AFRICAN HORSE SICKNESS VIRUS NON-STRUCTURAL PROTEIN NS5

Nhlakanipho Mtambo^{1*}, Eshchar Mizrahi¹, Vida van Staden¹

¹Department of Biochemistry, Genetics and Microbiology, *nlpho.mtambo@up.ac.za

African horse sickness virus (AHSV) is an economically important orbivirus affecting equids throughout sub-Saharan Africa. The virus has a segmented double-stranded RNA (dsRNA) genome. Genome segment 10 (Seg-10) encodes the well-characterised NS3 protein, involved in viral egress. It also contains an alternative +1 open reading frame (ORF) predicted to encode the non-structural protein NS5. Despite preliminary evidence of NS5 expression and a proposed role in host-virus interactions, its intracellular distribution remains poorly characterised. This study aimed to investigate the subcellular localisation and targeting motifs of AHSV NS5. To achieve this, AHSV Seg-10 nucleotide sequences were retrieved from NCBI GenBank and analysed, yielding a dataset of 105 unique NS5 ORFs. In silico bioinformatic analysis predicted conserved N-terminal features associated with mitochondrial targeting, alongside a putative nuclear localisation signal. Further, mutagenesis analysis demonstrated that N-terminal truncation or the substitution of arginine residues at positions 21 and 22 reduced predicted mitochondrial targeting scores across all sequences. To experimentally investigate this, recombinant baculovirus constructs encoding wild-type and mutant NS5 fused to a Strep-tag, a Twin-Strep-tag, or eGFP were expressed in Sf9 insect cells and subsequently visualised. Confocal microscopy revealed distinct subcellular distributions: Strep-tagged and Twin-Strep-tagged NS5 displayed predominant nuclear localisation, while eGFP fusions exhibited mixed nuclear and partial mitochondrial localisation. These findings align with the bioinformatic predictions for dual organelle localisation and suggest the potential involvement of N-terminal arginine residues in mitochondrial targeting. Overall, these results provide a foundation for future functional characterisation of NS5 in the context of AHSV infection. Understanding these host-virus interactions may ultimately aid in developing novel antiviral strategies.

GENETIC DIVERSITY OF EHRLICHIA RUMINANTIIUM BASED ON MULTILOCUS SEQUENCE TYPING OF ISOLATES FROM INDIGENOUS GOATS AND AMBLYOMMA HEBRAEUM TICKS

Xolile Nuse^{1,2*}, H.C. Steyn⁴, M.A. Van Der Nest³, K. Hadebe², F.C. Muchadeyi², E.F. Dzomba¹

¹Discipline of Genetics, School of Life Sciences, University of KwaZulu-Natal, South Africa. ²Agricultural Research Council - Biotechnology Platform, Onderstepoort, Pretoria, South Africa. ³Department of Biochemistry, Genetics and Microbiology, University of Pretoria, Pretoria, South Africa. ⁴Department of Molecular Biology, Agricultural Research Council - OVR, Onderstepoort 0110, South Africa. *xolilenuse@gmail.com

Heartwater, an infectious tick-borne disease caused by the intracellular bacterium *Ehrlichia ruminantium* and transmitted by *Amblyomma* ticks, remains a major constraint on ruminant production in sub-Saharan Africa due to high mortality. The main challenge in controlling the disease is the lack of a broadly effective vaccine, which is thought to be related to the high genetic variation among pathogen strains circulating in the same area. This study investigated the genetic diversity and phylogenetic structure of *E. ruminantium* isolates circulating in South African indigenous goats and *A. hebraeum* ticks using multilocus sequence typing (MLST). A total of 381 goat blood samples and 70 tick samples were screened using pCS20 quantitative real-time PCR, yielding 56 (14.7%) and 31 (44.3%) positive samples, respectively. Positive samples were characterized by sequencing four loci (*glta*, *groEL*, *sucA2*, and *pCS20*), followed by locus-specific and concatenated phylogenetic analyses. MLST identified 11 unique sequence types among 44 sequences and a haplotype diversity of 0.77, demonstrating considerable genetic heterogeneity within the pathogen population. Phylogenetic analyses revealed multiple co-circulating lineages, including clusters related to reference strains such as Gardel, Kümm2, and Grootvallei, alongside divergent genotypes suggestive of locally evolving variants. Population genetic analyses indicated weak host-associated structuring between goat- and tick-derived isolates ($F_{ST} = 0.0247$; $p > 0.05$), while AMOVA showed that 97% of genetic variation occurred within host groups. Haplotype network and multivariate analyses further supported a shared goat-tick transmission system, with no evidence of strict host specificity. Selection analyses suggested that adaptive evolutionary processes may contribute to the observed diversity. These findings demonstrate that *E. ruminantium* populations in South Africa are highly diverse and weakly structured by host, providing important insights into pathogen evolution, transmission dynamics, and future heartwater vaccine development.

Keywords: *E. ruminantium*, heartwater, multilocus sequence typing, genetic diversity, phylogenetic, indigenous goats, *A. hebraeum*

GENOMIC SURVEILLANCE OF TREHALOSE-6-PHOSPHATE SYNTHASE IN ANOPHELES GAMBIAE: A PROMISING TARGET FOR NEXT-GENERATION INSECTICIDES

Mercy Bella-Omunagbe^{1,2*}, Nina Ghislaine Yensii^{1,2}, Abiodun H. Adebayo¹, Shalom N. Chinedu¹

¹Department of Biochemistry, Covenant University, Ota. ²Covenant Applied Informatics and Communications - Africa Centre of Excellence, Covenant University Ota. *mercybella.omunagbe@stu.cu.edu.ng.

Background: The increasing prevalence of insecticide resistance in *Anopheles gambiae* threatens malaria control efforts across Africa, highlighting the need for novel molecular targets. Trehalose-6-phosphate synthase (TPS), a key enzyme in trehalose biosynthesis, is essential for survival and development and is absent in humans, making it a promising insecticide target. Its genetic variability and evolutionary stability in natural populations remain unclear. **Methods:** We analysed whole-genome sequence data from the Ag1000G Phase 3.0 dataset, comprising populations of the *Anopheles gambiae* complex from 19 African countries. The SNPs and amino acid changes allele frequencies, nucleotide diversity, and Tajima's D were analysed using the malariagen_data toolkit. Analyses focused on non-synonymous variants with allele frequency ≥ 0.05 in at least one cohort. Linkage disequilibrium was assessed using Lewontin's D' method. Functional impacts were predicted using SIFT and PROVEAN. **Results:** A total of 4590 variants were identified in TPS, including 459 non-synonymous mutations, of which 19 occurred at appreciable frequencies. These variants were found to be generally rare, geographically restricted and species-specific, and to have no mutations shared across all species. LD analysis showed the variants were either in complete linkage or not linked. Functional predictions identified 32% of variants as deleterious, with high-confidence mutations restricted to *Anopheles bwani* populations in Tanzania. Low nucleotide diversity ($\pi < 0.01$) and predominantly negative Tajima's D indicate strong purifying selection and functional constraint. Conclusion: TPS showed low genetic diversity, limited functional variation and strong signatures of purifying selection across the *Anopheles gambiae* complex. These suggest that TPS represents a genetically stable molecular target for vector control interventions, although further functional and experimental studies are required to validate it.

ISOLATION AND CHARACTERIZATION OF BIOCIDES PRODUCING FUNGUS AGAINST ANOPHELES MOSQUITOES FROM PESTICIDE FACTORY EFFLUENTS

Tariku Feyisa Wakjira^{*}, Musin Keel Abas^{1,2}, Misrak Kebede^{1,2}, kejelcha Fikadu^{3,4}, Tirsit Tibebe⁴

¹Addis Ababa science and technology university. ²Biotechnology Department, Addis Ababa science and technology university. ³Biotechnology Department, Dembi Dollo university. ⁴Holota National Agricultural Biotechnology Research Centre. *tariku2918@gmail.com.

Background: Malaria diseases has become global health challenge due to mosquito resistance to insecticide. Therefore, primary goal of this study was to isolate, characterize and evaluate fungal biocides from pesticide-contaminated environments for their potential use as eco-friendly mosquito control agents. The study insights analyse the metabolic profiles of fungal isolates and the genetic diversity of strains that show more efficiency and the ability to produce potent secondary metabolites.

Methodology: Pesticide factory effluent sample was used to isolate and characterized fungal strain using morphological, biochemical and molecular characterizations. The isolates were screened for spore production and larvicidal activities. The spores extracted from the isolates were evaluated using GC-MS. Molecular Docking was used to identify the interaction of the identified compound with target molecule. **Result:** A total of thirty fungal isolates exhibiting distinctive morphological and biochemical characteristics were obtained. Among them, two isolates identified as W1 and W4 demonstrated strong potential. These findings showed the potentials of pesticide adapted fungi as a source of ecofriendly mosquito control with different doses of fungal spore suspension resulted in mosquito larval mortality rate ranging from 6.67% to 100%. Molecular and phylogenetic analysis indicated that isolate W1 showed 99% similarity to the species *Penicillium crustosum*, while isolate W4 was found to have 100% similarity to the species *Rhizopus arrhizus* in the NCBI database. The chromatograms analysis of W1 and W4 revealed the presence of 33 and 29 chemical compounds, respectively, while the molecular Docking studies predicted strong binding affinity of several compounds, particularly Phenol, 2-(1-phenylethyl) - aligning with recognized insecticide targets. Conclusion: The result of this study suggest that the fungal isolates identified could be a promising candidate as a biocontrol agent for mosquito larvae and adults with optimization of the culture condition and formulation studies.

Keywords: biocide, fungi, mosquito, larvae, pesticide

GRAPH NEURAL NETWORK-BASED DISCOVERY OF MIRNA-MRNA REGULATORY INTERACTIONS IN BREAST CANCER

Nothando Gama^{1*}, Hocine Bendou¹

¹South African National Bioinformatics Institute (SANBI), SA Medical Research Council Bioinformatics Unit, University of the Western Cape, Private Bag X17, Bellville, Cape Town 7535, South Africa. *nothandograce82@gmail.com.

Breast cancer remains one of the leading causes of cancer-related mortality among women worldwide, with dysregulation of microRNA (miRNA)-mediated gene regulation contributing to tumor initiation, progression, and metastasis. MicroRNAs are small non-coding RNAs that regulate gene expression through post-transcriptional repression of target messenger RNAs (mRNAs), making the identification of miRNA-mRNA interactions critical for understanding disease mechanisms and discovering therapeutic targets. Despite advances in computational prediction, many biologically relevant miRNA-mRNA interactions remain unidentified. Heterogeneous graph-based deep learning provides an opportunity to model complex regulatory networks and identify novel interactions. In this study, we developed a Heterogeneous Graph Transformer (HGT)-based link prediction framework to discover miRNA-mRNA regulatory interactions associated with breast cancer. Experimentally validated interactions from miRTarBase were integrated with transcriptomic data from TCGA breast invasive carcinoma (TCGA-BRCA). To enhance network coverage, ovarian cancer data (TCGA-OV) were incorporated, leveraging shared regulatory mechanisms between the two cancers. A heterogeneous graph comprising miRNA and mRNA nodes was constructed using mean expression levels and node degree as node features. The model was trained using a threshold-optimized negative sampling strategy and evaluated through link prediction. The HGT framework achieved an accuracy of 82%, AUC-ROC of 90%, AUC-PRC of 80%, F1-score of 75%, and Matthews Correlation Coefficient (MCC) of 62%. High-confidence novel predictions included hsa-miR-4775 targeting SMAD4, ABCA1, and ZMAT3, and hsa-miR-520d-5p targeting BMP2. These targets are implicated in TGF- β signalling, epithelial-mesenchymal transition, tumour suppression, and p53-mediated apoptosis implicated in breast cancer progression. Overall, this study demonstrates the potential of graph-based deep learning to accelerate the discovery of miRNA-mediated regulatory mechanisms and prioritize candidate interactions for experimental validation in breast cancer.

RECONSTRUCTING CERVICAL CANCER TUMOUR MICROENVIRONMENT STATES FROM DNA METHYLATION PROFILES USING A MACHINE LEARNING FRAMEWORK

Saltil Hamese^{*}, Mutsha Takundwa¹, Earl Prinsloo², Deepak Balaji Thimiri Govinda Raj¹

¹Synthetic Nanobiotechnology and Biomachines, Synthetic Biology and Precision Medicine Centre, CSIR Pretoria, South Africa. ²Biotechnology Innovation Centre, PO Box 94, Rhodes University, Makhanda, 6146 South Africa. *hamesesaltil@gmail.com.

Background: Cervical cancer remains a major cause of cancer-related morbidity and mortality globally, particularly in low- and middle-income countries where the burden is exacerbated by persistent human papillomavirus (HPV) infection and high rates of HIV co-infection. Despite the availability of large-scale genomic, epigenomic, transcriptomic, and proteomic datasets, translating these data into biologically meaningful tumour classifications and clinically actionable insights remains challenging. A major limitation is the lack of computational frameworks capable of reconstructing tumour microenvironment (TME) states from bulk patient datasets and systematically linking these states to representative experimental model systems. **Methods:** We developed a DNA methylation-based machine learning framework for tumour microenvironment state inference and molecular stratification in cervical cancer. The framework integrates epigenomic feature selection, methylation-based deconvolution, unsupervised clustering, and predictive modelling to reconstruct biologically meaningful tumour states directly from patient DNA methylation profiles. By incorporating signals derived from malignant epithelial cells, immune populations, stromal compartments, and host-associated biological processes, the approach enables comprehensive characterisation of cervical tumour ecosystems. **Results:** Application of the framework to cervical cancer cohorts identified four distinct tumour microenvironment states characterised by differences in immune activity, stromal composition, epithelial differentiation, and proliferative signalling. These reconstructed states revealed substantial inter-tumour heterogeneity and provided insight into the biological consequences of HPV-driven carcinogenesis and HIV-associated immune modulation. Notably, HIV-associated molecular perturbations were largely restricted to immune-inflamed tumours, indicating that HIV influences specific biological contexts rather than defining a universal cervical cancer subtype. The inferred tumour states also enabled systematic alignment of patient tumours with cervical cancer cell line models, facilitating identification of experimental systems that more accurately recapitulate patient-derived phenotypes. **Conclusions:** This framework provides a reproducible approach for methylation-driven tumour microenvironment reconstruction and molecular stratification in cervical cancer. By linking patient tumour states to experimentally tractable models, it supports biomarker discovery, disease-risk stratification, therapeutic vulnerability assessment, and precision oncology research. More broadly, the framework demonstrates how machine learning applied to epigenomic data can bridge the gap between patient biology and preclinical experimentation, supporting future advances in cervical cancer prevention, screening, and treatment.

SURVEILLANCE AND MONITORING OF ANTIMICROBIAL-RESISTANT HUMAN PATHOGENIC AND ZOO NOTIC BACTERIA IN THE NGAMILAND AND OKAVANGO DELTA

Patience Motshosi^{1*}, Kabo Baruti²

¹Botswana Harvard Health Partnership, Gaborone, Botswana. ²Okavango Research Institute, University of Botswana, Maun, Botswana. *natiencemotshosi@gmail.com

Antimicrobial resistance (AMR) represents a critical global health threat, with projections of 10 million annual deaths by 2050, disproportionately affecting low- and middle-income countries (LMICs). Freshwater environments at high human-animal-wildlife interfaces, such as the Okavango Delta and Ngamiland, serve as key reservoirs and dissemination pathways for pathogenic bacteria and AMR genes. This environmental surveillance study collected surface water samples from eight sites: three along the Thamalakane River in Maun and four semi-permanent pools within Moremi Game Reserve. Physicochemical parameters were recorded at river sites only. Samples were concentrated by centrifugation, and total nucleic acids were extracted and sequenced using the Illumina NextSeq2000 platform. Taxonomic classification employed Kraken2, while AMRFinder identified antibiotic resistance genes (ARGs). Statistical analyses included independent t-tests, one-way ANOVA, Shannon diversity index, and PERMANOVA for bacterial alpha diversity comparisons. Seasonal variation significantly influenced salinity and electrical conductivity ($p = 0.0007$ and $p = 0.0001$, respectively), while temperature, pH, and culturable bacterial counts remained unaffected ($p > 0.05$). No significant inter-site differences were detected across any parameter. Metagenomic sequencing identified *Acinetobacter lwoffii* and *Acinetobacter johnsonii* (~1–3%) as the most abundant pathogens. Clinically significant zoonotic pathogens, including *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Stenotrophomonas maltophilia*, and *Proteus mirabilis*, were detected at low abundance. The resistome was dominated by efflux pump genes *MexF* (18.3%) and *MexB* (8.3%), alongside the clinically relevant β -lactamase TEM-116 (17.4%). Hydrological variability primarily influenced ionic composition, while bacterial load remained consistent across sites and seasons. The co-detection of TEM-116 and transposase gene *TnIB* raises concern for horizontal gene transfer (HGT) between environmental and clinical bacteria. Even low-abundance detection of *A. baumannii*, a known HGT facilitator, is clinically significant. These findings underscore the value of freshwater metagenomic surveillance as an early warning system for AMR in Botswana and comparable LMICs.

ASSESSING THE GENETIC DIVERSITY AND STRUCTURE OF SPOTTED HYENA (*CROCUTA CROCUTA*) POPULATIONS ACROSS AFRICA.

Kagisho Dibakoane*, Yoshan Moodley¹, Michael Westbury².

¹Department of Zoology, University of Venda, Thohoyandou, South Africa. ²Technical University of Denmark. *kagisho@univen.ac.za

The spotted hyena (*Crocota crocuta*) is a socially complex, ecologically important carnivore whose range-wide genomic variation remains poorly characterised. We analysed 55 individuals from across sub-Saharan Africa using whole-genome and restriction site-associated DNA (RAD-seq) data to quantify genetic diversity, infer population structure, and assess gene flow. Heterozygosity ranged from 0.1485 to 0.3688 among populations. Population genomic analyses revealed three major clusters corresponding to western, eastern, and southern Africa, with finer-scale subdivision into five regional groups. D-statistics indicated historical and ongoing gene flow, particularly between eastern and southern populations, while the western cluster was more differentiated and exhibited the lowest genetic diversity. These patterns parallel those observed in other large African carnivores and likely reflect a combination of historical biogeographic processes and recent anthropogenic fragmentation. Our results provide the most geographically comprehensive genomic assessment of spotted hyenas to date and highlight the need for targeted conservation strategies that prioritise restoring connectivity to western populations.

TEMPORAL DYNAMICS OF THE GUT MICROBIOME DURING SMOLTIFICATION IN ATLANTIC SALMON (*SALMO SALAR*)

Badroddin M¹, Difford G², Rhode C¹.

¹Department of Genetics, Stellenbosch University, Stellenbosch. ²Institute of Animal and Aquaculture Sciences, Norwegian University of Life Sciences, Ås, Norway. *23563636@sun.ac.za

Smoltification is a critical developmental transition in Atlantic salmon (*Salmo salar*) that enables seawater adaptation. Although the gut microbiome is increasingly recognized as important in host development and metabolism, its temporal dynamics during smoltification remain poorly understood. This study aimed to investigate temporal changes in the gut microbiome of Atlantic salmon ($n = 162$) across the smoltification period. Gut microbial communities were characterized through 16S rRNA gene sequencing followed by DADA2 preprocessing and Phyloseq-based downstream analyses to assess community composition, diversity and temporal variation. Temporal restructuring of the gut microbiome was observed. Alpha diversity varied significantly among sampling weeks with Chao1 richness ranging from 80-200 taxa (Kruskal-Wallis $\chi^2 = 43.95$, $df = 7$, $p = 2.19 \times 10^{-7}$), Simpson diversity exceeding 0.75 ($\chi^2 = 23.75$, $df = 7$, $p = 0.00126$) and Shannon diversity ranging from 2-5 ($\chi^2 = 26.954$, $df = 7$, $p = 0.00034$). Beta diversity demonstrated significant temporal differentiation in community composition for Bray-Curtis (PERMANOVA: $R^2 = 0.073$, $F = 12.615$, $p = 0.001$), suggesting ecological succession rather than a uniform increase in microbial diversity. The microbiome included groups such as Bacillota and Pseudomonadota, which may serve as candidate microbial indicators of seawater adaptation. These findings indicate that gut microbiome composition is a complex quantitative trait shaped by host developmental state and environmental conditions. Improved understanding of microbiome dynamics during smoltification may contribute to strategies to optimize salmon health and welfare at sea.

UNRAVELLING THE ELUSIVE MATING STRATEGY OF DEMATOPHORA NECATRIX: A COMBINED GENOMIC AND PHYSIOLOGICAL APPROACH

Annabel Norval^{1,2*}, Markus Wilken¹, Noëliani van den Berg^{1,2}

¹Forestry and Agricultural Biotechnology Institute (FABI), Department of Biochemistry, Genetics and Microbiology, University of Pretoria. ²The Avocado Research Programme, Forestry and Agricultural Biotechnology Institute (FABI), University of Pretoria. *annabel.norval@un.ac.za

Sexual reproduction in pathogenic ascomycete fungi generates genetic variation, accelerating natural selection and contributing to fungicide resistance and the ability to overcome host resistance. This process is governed by the mating-type (MAT1) locus, comprising two idiomorphs (MAT1-1 and MAT1-2), each encoding transcription factors regulating downstream sexual development genes. The genes within each idiomorph vary considerably, forming the genetic basis distinguishing heterothallic from homothallic fungi. Mating-type is conventionally determined through *in silico* BLAST-based searches or *in vitro* crossing experiments. For the avocado pathogen, *Dematophora necatrix*, both approaches have been challenging. Canonical MAT1 genes appear entirely absent across sequenced Xylariales genomes, and no crossing experiments have, to date, characterised mating strategy of *D. necatrix*. This study employed a dual genomic-physiological approach to elucidate this mating strategy. Full-genome annotation of Xylariales species using the funannotate pipeline, incorporating RNA-seq data, enabled comparative identification of candidate mating-type genes through non-conventional approaches. Candidate proteins were subjected to HMMER searches against a custom mating-type protein database, InterProScan domain analysis, maximum likelihood phylogenetic reconstruction, and three-dimensional structural prediction via AlphaFold3, with structural comparisons quantified using RMSD calculations in PyMOL — a method not previously applied to mating-type gene identification. Concurrently, controlled crossing experiments between eight single-ascospore isolates, derived from field-collected sexual structures and subjected to Illumina whole-genome sequencing, assessed mating-type segregation physiologically. Together, these approaches yielded candidate mating-type genes via a novel methodology, provided genome-wide annotations for the Xylariales, and generated foundational insights into the mating strategy of *D. necatrix*, aiding downstream control strategies of this devastating pathogen.

ADVANCING AVIAN GENOMICS: A DE NOVO ASSEMBLY PIPELINE TO GENERATE A HIGH-QUALITY REFERENCE GENOME FOR THE SOUTHERN BALD IBIS (*GERONOTICUS CALVUS*)

Ashley Mortimer¹, Anri van Wyk¹, Arrie Kloppe¹, Paulette Bloomer¹

¹Molecular Ecology and Evolution Programme, Department of Biochemistry, Genetics and Microbiology, University of Pretoria. *u21536794@tubs.co.za

The Southern Bald Ibis (*Geronticus calvus*) is a near threatened southern African avian species of special interest, displaying behaviour vastly differing from other ibis species; apart from its closest relative, the Northern Bald Ibis (*Geronticus eremita*). Due to their unique preference for both high rainfall and altitude, their distribution is restricted to parts of Lesotho, eSwatini and South Africa. Genetically the species has received minimal attention, with a single mitochondrial DNA study comparing the congeners, of which neither has an available reference genome. A reference genome will be the first step in aiding the species' conservation, acting as a guide for future research, including population genetics/genomics, landscape genomics, and providing a baseline for investigating potential adaptation within the species due to their unique habitat preferences. Oxford Nanopore Technology (ONT) long-read sequencing was performed to generate a high accuracy reference genome. The following de novo assembly pipeline was utilised: trimming, filtering, assembly, raw read mapping, polishing, haplotig purging and scaffolding; with quality checks in-between. A final genome assembly of 1.218 Gb was produced, consisting of 464 contigs, with an N50 value of 25,9698 Mb, only containing 20 ambiguous bases and a 99.4% BUSCO completeness score. Based on available avian reference genomes, the genome produced exhibits higher BUSCO completeness scores, lower scaffold N50 values and the expected 1.2 Gb genome size of avian species. The assembly pipeline generated a reference-level genome for the species to be utilized in future population genetics and genomics studies, allowing for informed conservation efforts.

AN IMMUNOINFORMATICS APPROACH FOR CROSS-SPECIES DIAGNOSTICS: VALIDATING A *FASCIOLA GIGANTICA* SURROGATE ANTIGEN FOR *SCHISTOSOMA HAEMATOBIIUM* DETECTION

Adeyemi O. Adesida¹, Kabirat A. Suleiman², Stella A. Kayode⁴, Oyetunde T. Oyetunde²

¹Institute of Genomics and Global Health, Redeemer's University, Ede, Osun State, Nigeria. ²Department of Biosciences and Biotechnology, University of Medical Sciences, Ondo, Ondo State, Nigeria. ³Department of Science Laboratory Technology, University of Medical Sciences, Ondo, Ondo State, Nigeria. ⁴Centre for Molecular Biosciences and Medical Genomics, University of Medical Sciences, Ondo, Ondo State, Nigeria. *adesida19852@run.edu.ng

Background: Mass drug administration for urogenital schistosomiasis requires the support of improved, precision-based diagnostic surveillance. Sourcing *Schistosoma haematobium* antigens for rapid diagnostic tests (RDTs) presents significant challenges. Identifying a highly conserved, cross-reactive structural epitope in livestock flukes like *Fasciola gigantica* offers a viable pathway to mass-producing surrogate biomarkers for precision diagnosis. **Methods:** An immunoinformatics pipeline was employed, beginning with a proteome-wide reciprocal BLASTp ortholog identification between *S. haematobium* and *F. gigantica*. Candidates were computationally filtered for secretory signal peptides and extracellular surface accessibility. To prevent false positives, targets were subjected to a strict negative selection screen against the human reference proteome. The 3D structural viability of the final surrogate was tested via molecular docking (HADDOCK 2.4) against the CAMPATH-1H antibody and thermodynamic validation via the MM/GBSA method (HawkDock). **Results:** The pipeline identified 583 orthologous pairs sharing >80% sequence identity, ultimately isolating a single "SAFE" pair: MS3_00011174.1 and FGIG_11052, which returned 0% sequence homology to human proteins. Functional annotation mapped the targets to the Ly-6/uPAR/CD59 glycoprotein family (Snake toxin-like superfamily), characterized by a highly stable, disulfide-linked "three-finger" fold. Conformational epitope mapping revealed a perfectly conserved 23-amino-acid structural core (CQREWYMDWRCVECCGELCNYY) situated within this extracellular domain. Molecular docking confirmed high-affinity binding to the CAMPATH-1H antibody, yielding HADDOCK scores of -94.1 ± 3.8 and -98.0 ± 3.5 for the *S. haematobium* and *F. gigantica* targets, respectively. MM/GBSA calculations verified spontaneous and stable interactions in a solvent environment, with binding free energies of -75.77 kcal/mol and -83.28 kcal/mol. **Conclusion:** This immunoinformatics framework computationally validates the conserved G-A-Y-C-Q motif and C-terminal domain as highly accessible, broad-spectrum biomarkers. This *Fasciola* surrogate target provides a biologically robust foundation for developing next-generation rapid diagnostic tests free of host autoantibody cross-reactivity.

DOES MULTI-TASK SUPERVISION TEACH A CLASSIFIER WHERE TO LOOK? A QUANTITATIVE ATTENTION-LOCALIZATION STUDY IN BRAIN TUMOR MRI

Tisetso Letuka*, Amukelani Mtshabi¹

¹Ezintsha, Wits Health Consortium, Faculty of Health Sciences, University of the Witwatersrand. *tisetsoletuka@gmail.com

Background: Explainability methods, such as Grad-CAM, are widely trusted to reveal where an artificial intelligence model focuses when making a medical diagnosis. Multi-task learning is frequently proposed as a strategy to encourage models to attend to clinically relevant regions. However, whether explicit spatial training in a dedicated task head actually improves the internal attention of a shared classification head remains largely unproven in medical imaging. This study quantitatively evaluates whether a lesion-localization head aligns a model's diagnostic attention maps with true, radiologist-annotated tumor centers across 30 pathology classes. **Methodology:** We evaluated a multi-task deep learning architecture (EfficientNet-B4 backbone, trained from scratch) trained on identical brain MRI data to simultaneously classify pathologies and predict (x,y) lesion coordinates. Using a 1,128-image validation set encompassing 30 distinct tumor types and normal controls, we generated post-hoc Grad-CAM++ attention heatmaps for correctly classified abnormal images. We calculated the Euclidean distance between the heatmap's weighted centroid and the true radiologist-annotated tumor locations, comparing results between a multi-task model (loss weight $\beta=0.5$) and a single-task classifier ($\beta=0.0$). **Results:** Both architectures achieved high diagnostic performance (multitask macro-F1: 0.91; single-task: 0.91). The multi-task model's dedicated localization head demonstrated strong spatial accuracy, achieving a median error of 30.7 pixels, with 85.5% of predictions falling within 64 pixels of the true lesion center. However, despite this internal accuracy, its classification attention maps remained highly misaligned, showing a median error of 227.4 pixels (only 2.0% within 64 pixels). Crucially, the single-task model showed a statistically indistinguishable attention profile (226.8 pixel median error; 1.9% within 64 pixels), despite receiving zero spatial training. **Conclusion:** Explicitly teaching a model to locate a lesion yields highly accurate coordinate predictions but does not meaningfully improve the spatial alignment of its classification attention, suggesting that diagnostic decisions may rely on features that are partially decoupled from localized tumor boundaries. This has direct implications for clinical trust: post-hoc attention maps cannot be blindly relied upon for diagnostic justification. Instead, clinical decision support tools should prioritize explicitly supervised, dedicated localization outputs over passive explainability heatmaps.

THE ROLE OF THE GUT MICROBIOME IN INFLAMMATION AND MULTIMORBIDITY IN AFRICAN POPULATIONS: EVIDENCE FROM AWI-GEN

Aaron Y. Berkman^{1,2}, Luicer A. Olubayo^{1,2}, Scott Hazelhurst^{1,3}

¹Sydney Brenner Institute for Molecular Bioscience, Faculty of Health Sciences, University of the Witwatersrand. ²Division of Human Genetics, School of Pathology, Faculty of Health Sciences, University of the Witwatersrand. ³School of Electrical and Information Engineering, Faculty of Engineering, University of the Witwatersrand. *183165@students.wits.ac.za

The gut microbiome is a central regulator of systemic inflammation, a pathway that links diet and environment to multimorbidity — the accumulation of chronic disease. This project aims to determine how gut-microbial function relates to inflammation and multimorbidity in African populations and to resolve the strain-level genomic basis of any associations. C-reactive protein (CRP) offers an interpretable marker of the inflammatory load that drives it. Multimorbidity is an escalating burden across sub-Saharan Africa, yet the evidence base remains overwhelmingly Western. African populations are critically under-represented despite their distinct diets, infectious exposures, and microbial ecology, and the reference databases on which standard functional profilers depend largely exclude novel African gut taxa. Quantifying these organisms, therefore, calls for de novo assembly rather than reference-based methods. The project is nested within the AWI-Gen 2 Microbiome Project, the largest African population-representative metagenomic resource to date (1,801 participants from Burkina Faso, Ghana, Kenya, and South Africa); it pairs deep shotgun (n=1,801) and long-read (n=209) metagenomic sequencing with serum CRP and harmonised multimorbidity data, offering a rare setting in which these relationships can be properly resolved. To date, metagenome-assembled genomes have been constructed de novo and their gene-level functional contributions quantified, recovering taxa that reference methods miss. A pre-specified panel of microbial functions, spanning short-chain fatty acid, bile-acid, and tryptophan-indole pathways, has been tested for association with CRP and multimorbidity. Building on this, a long-read pangenome workflow will resolve within-species structural variation in prioritised taxa, asking what strain-level genomic differences reveal about host-phenotype associations beyond community-level measures. We will present these reproducible Nextflow workflows together with preliminary findings from their application to the AWI-Gen dataset.

IMPACT OF REFERENCE GENOME, ALIGNMENT STRATEGY, AND QUANTIFICATION METHOD ON LOCUS-SPECIFIC HUMAN ENDOGENOUS RETROVIRUS (HERV) DETECTION DURING MONOCYTE-TO-MACROPHAGE DIFFERENTIATION

Tebogo Phago^{*}, Nikki Gentle¹, Vanessa Meyer²

¹Functional Genomics and Immunogenetics Research Group, School of Molecular and Cell Biology, Faculty of Science, University of the Witwatersrand. ²ImmunOmics Research Group, School of Molecular and Cell Biology, Faculty of Science, University of the Witwatersrand. *24362923@students.wits.ac.za

Human endogenous retroviruses (HERVs) are ancient proviral sequences that comprise approximately 8% of the human genome and have been increasingly implicated in cellular differentiation, immune regulation, and disease. However, their repetitive and dispersed genomic nature complicates accurate locus-specific expression analysis, and no standardized workflow currently exists for HERV profiling. This study aimed to identify HERV loci associated with monocyte-to-macrophage differentiation while evaluating how genome assembly, alignment strategy, and quantification method influence HERV detection. THP-1 monocytic cells were used for pipeline development, while U937 cells are being analysed as an independent validation model. RNA-sequencing reads were aligned to the GRCh38 and CHM13 reference genome assemblies using HISAT2 and STAR, followed by locus-level HERV quantification using TE-locator and MAJEC, among the few established tools for locus-specific transposable element expression analysis. Differential expression analysis was performed using DESeq2 (FDR < 0.05, log2 fold change ≥ 1). Overall mapping rates were comparable across assemblies and aligners (approximately 89–93%), although HISAT2 consistently produced a higher proportion of uniquely mapped reads and reduced multimapping relative to STAR. Downstream quantification was strongly influenced by alignment strategy, with HISAT2 yielding higher read assignment to annotated features (approximately 25–40 million) compared to STAR (15–25 million; $p = 0.0341$), indicating substantial pipeline-dependent variation in feature-level quantification. Differential expression results also varied across workflows, with CHM13 consistently identifying more differentially expressed HERV loci than GRCh38, highlighting the impact of reference genome choice on transposable element discovery. Pipeline-specific detection was observed, including over 571 loci uniquely identified by the HISAT2-TE-locator workflow, while a smaller subset of loci ($n = 253$) was consistently detected across all pipelines. Notably, a differentially expressed HERV located near the IFNB1 promoter was identified, suggesting a potential regulatory relationship with interferon signalling during macrophage differentiation. Overall, these results suggest that genome assembly, alignment strategy, and quantification method substantially influence locus-specific HERV detection and downstream biological interpretation, and findings are still being validated.

INTEGRATED PHENOTYPIC PROFILING AND THERAPEUTIC TARGET PRIORITIZATION OF CLINICAL MULTIDRUG-RESISTANT PSEUDOMONAS AERUGINOSA EXHIBITING BIOFILM-DRIVEN PERSISTENCE

Sinethemba Yakobi¹, Nontuthuko Maningi¹

¹School of Life Sciences, College of Agriculture, Engineering and Science, University of KwaZulu-Natal. *shyakobi@gmail.com

Multidrug-resistant (MDR) *Pseudomonas aeruginosa* poses a major clinical threat due to its capacity to form biofilms that promote persistence and therapeutic failure. In this study, we integrated phenotypic profiling of clinical isolates with structure-based computational analysis to identify and prioritize therapeutic targets associated with biofilm-driven persistence. Fifteen clinical isolates were characterized for antimicrobial susceptibility and biofilm-forming capacity, revealing a high prevalence of strong biofilm phenotypes across resistance classes and a significant positive association between resistance burden and biofilm biomass. A candidate persistence-associated target, PA0328 (AaaA), was structurally modelled and subjected to consensus binding pocket identification. Ensemble molecular docking of eight candidate ligands identified pimozone and epigallocatechin gallate (EGCG) as top-ranked compounds, with high reproducibility across replicate runs. Interaction topology analysis revealed distinct stabilization mechanisms, with pimozone exhibiting hydrophobic and π - π stacking-driven binding, and EGCG forming extensive hydrogen-bond networks. Cross-platform docking validation confirmed consistent binding site localization and ligand ranking. ADMET profiling showed contrasting translational profiles; pimozone displayed favourable drug-like properties and bioavailability, supporting its potential for repurposing, whereas EGCG exhibited strong binding stability but limited pharmacokinetic feasibility due to high polarity and poor absorption. This study establishes a scalable, simulation-independent framework for therapeutic target prioritization in MDR *P. aeruginosa*, integrating phenotypic, structural, and pharmacokinetic dimensions. The findings highlight PA0328 as a promising target and support prioritized repurposing candidate for further investigation.

BENCHMARKING DNBSEQ-T7 SEQUENCING PERFORMANCE USING GENOME IN A BOTTLE REFERENCE GENOMES: IMPACT OF LIBRARY PREPARATION STRATEGIES

Maano Malima¹, Ansia van Coller^{1,2}, Setshaba Taukobong^{1,2}, Samira Ghoor¹, Nganea Nangambi¹, Enrico Roode¹, Martin Naicker¹, Victoria Cole¹, Brigitte Glanzmann^{1,3}, Craig Kinnear^{1,3}, Nadia Carstens^{1,4}

¹South African Medical Research Council (SAMRC) Genomics Platform, Cape Town, South Africa. ²South African Medical Research Council (SAMRC) Biomedical Research and Innovation Platform (BRIP), Cape Town, South Africa. ³SAMRC Centre for Tuberculosis Research, Division of Molecular Biology and Human Genetics, Faculty of Medicine and Health Sciences, Stellenbosch University, Cape Town, South Africa. ⁴Division of Human Genetics, National Health Laboratory Service and School of Pathology, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa. *maano.malima@mrc.ac.za

Background: Advances in sequencing technologies have improved the accuracy, throughput and completeness of human genome characterisation, enabling more reliable detection of genetic variation. Well-characterised reference genomes are essential for benchmarking sequencing platforms and bioinformatics pipelines. **Methods:** We present whole-genome sequencing datasets generated on the MGI DNBSEQ-T7 platform for the Ashkenazi Jewish trio reference samples from the Genome in a Bottle (GIAB) Consortium. Libraries were prepared using three distinct MGI workflows; PCR-free, FastFS DNA, and Universal DNA followed by sequencing to a minimum of 400 million paired-end reads per sample ($\geq 30\times$ mean genome coverage). Raw reads were processed with a standardised GATK bioinformatics pipeline, and sequencing performance was evaluated against the GIAB high-confidence benchmark variant sets. **Results:** All three workflows produced high-quality data with strong overall concordance to the GIAB truth sets. The PCR-free library preparation demonstrated the best indel-calling performance and the lowest Mendelian violation rates across the Ashkenazi trio. **Conclusion:** This publicly available dataset provides a valuable resource for benchmarking DNBSEQ-T7 sequencing performance and for assessing the influence of different library preparation strategies on whole-genome variant detection. The results support the adoption of the DNBSEQ-T7 platform in high-throughput genomic research and reinforce its utility for producing accurate variant calls in precision-medicine and bioinformatics pipelines relevant to African genomic diversity studies.

Keywords: DNBSEQ-T7; whole-genome sequencing; Genome in a Bottle; library preparation; variant calling; bioinformatics; African genomics

THANK YOU TO OUR SPONSORS

DIPLOMICS



CONNECTING RESEARCHERS & WORLD CLASS OMICS FACILITIES



MEET THE ORGANISING COMMITTEE

Caivil Ndobela Chairperson

*President of SASBi Student Council,
University of Pretoria*

Joshua Sampson Chairperson

*President of SAGS Student Society,
Stellenbosch University*

Odireleng Mosuwe Program

*Treasurer of SASBi Student Council,
University of Pretoria*

Nozipho Magagula Program, Outreach & Finance

*Treasurer of SASBi Student Council,
University of the Witwatersrand*

Thulani Nkosi Finance

*Development Officer of SASBi Student
Council,
Stellenbosch University*

Sandile Ntloko Design & Outreach

*Deputy Development Officer of SASBi Student
Council,
University of Johannesburg*

Yonatan Wolberg Outreach

*Media Officer of SASBi Student Council,
University of Witswatersrand*

GOEBERHA TRAVEL CITY GUIDE



Welcome to the “Friendly City”

Port Elizabeth, officially renamed Gqerrha, is located on the breathtaking shores of Algoa Bay in the Eastern Cape Province. Known for its rich cultural history, pristine wind-swept coastlines, and exceptional wildlife access, it offers delegates a perfect blend of academic atmosphere and natural beauty.



Local Highlights and Attractions

The Beachfront Promenade:

Take a walk along Hobie Beach, HumeWood Beach or King's Beach right outside the central hub. Ideal for clearing your head between dense scientific tracks.

The Boardwalk Precinct:

Located in Summerstrand, this area features excellent dining, shopping, and entertainment walks surrounding an artificial lake.

Addo Elephant National Park:

Situated just 30 - 40 minutes outside the city, it is the third largest national park in South Africa and offers world-class viewing of the “Big Seven” in a malaria free environment.

Route 67:

A fascinating cultural arts heritage trail consisting of 67 public artworks symbolizing the 67 years Nelson Mandela dedicated to the freedom of South Africa.



Culinary Recommendations

Seafood:

Try the local catch (especially Kingklip or Calamari) at restaurants along the beachfront or near the harbor.

Coffee Shops:

Summerstrand and the nearby Walmer area host superb artisan cafes perfect for morning prep or informal meetings.

